

# Three Architectures

# Reshaping AI

**Cerebras · Mercury 2 · Qwen World**

*Speed. Diffusion. Simulation.*

# The Status Quo

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## How We Build AI Today

- GPU clusters — the default for a decade
- Autoregressive models — one token at a time
- Reactive agents — try, observe, try again

## The Problems

- **Latency** kills real-time applications
- **Sequential** generation caps throughput
- **Blind** agents can't think ahead

# Meet the Three Rebels

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## Cerebras

**Hardware  
Beast**

Wafer-scale  
chips —  
no inter-chip



## Mercury 2

**Diffusion  
Rebel**

Parallel text  
generation



## Qwen World

**The  
Simulator**

Predicts  
consequences  
before acting

# CEREBRAS

**The Hardware Beast**

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# The WSE-3 Wafer-Scale Engine

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57x

Larger than a standard GPU die

| Metric      | WSE-3                  | NVIDIA B200          |
|-------------|------------------------|----------------------|
| Die size    | 46,225 mm <sup>2</sup> | ~800 mm <sup>2</sup> |
| Transistors | 4 trillion             | 208 billion          |
| AI cores    | 900,000                | 18,432 CUDA          |
| Memory BW   | 20 PB/s                | ~4 TB/s (HBM3e)      |
| Peak perf   | 125 PF (FP16)          | —                    |

# Why Wafer-Scale Matters

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## GPU Cluster

```
graph LR; GPU1 <--> GPU2 <--> GPU3; GPU4 <--> GPU5 <--> GPU6; GPU1 <--> GPU4; GPU2 <--> GPU5; GPU3 <--> GPU6;
```

GPU<sub>1</sub> ↔ GPU<sub>2</sub> ↔ GPU<sub>3</sub>  
↕       ↕       ↕  
GPU<sub>4</sub> ↔ GPU<sub>5</sub> ↔ GPU<sub>6</sub>

- Data spends most time **moving**, not computing
- Interconnect is the bottleneck

## Wafer-Scale

```
graph LR; subgraph Die; direction TB; A[ONE MASSIVE DIE  
(model on-chip)]; end;
```

ONE MASSIVE DIE  
(model on-chip)

- Zero inter-chip hops
- Sub-second time-to-first-token

# The Yield Problem — Solved

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“ *"A single defect on a wafer would ruin the whole chip."* ”

**Cerebras solution:** Redundant cores — defective ones are bypassed.

This was the **key engineering breakthrough** that made wafer-scale viable.

# Cerebras: Key Numbers

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## Speed

**15× faster**  
inference vs  
GPUs

**1000+ tok/s** on  
large models



## Latency

**Sub-second**  
time-to-first-  
token

Real-time voice  
agents become  
viable



## Efficiency

Less energy  
wasted on data  
movement

One chip vs  
complex multi-  
GPU topologies



# MERCURY 2

The Diffusion Rebel

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# The Paradigm Shift

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## Autoregressive (GPT et al.)

Token<sub>1</sub> → Token<sub>2</sub> → Token<sub>3</sub> → Token<sub>4</sub>

- **Sequential** — each token depends on the previous
- ~50–100 tokens/sec
- Inherent latency floor

## Diffusion (Mercury 2)

Noise →  $\left[ \begin{array}{c} \text{Token}_1 \\ \text{Token}_2 \\ \text{Token}_3 \end{array} \right] \rightarrow \text{Refined text}$

- **Parallel** — multiple tokens generated simultaneously
- **1000+ tokens/sec**

# Mercury 2 Specs

| Capability   | Detail                                                   |
|--------------|----------------------------------------------------------|
| Speed        | 1000+ tok/s on standard GPUs                             |
| Latency      | ~3s → ~ <b>300ms</b> (reasoning)                         |
| Context      | 128K tokens                                              |
| API          | OpenAI-compatible (drop-in)                              |
| Capabilities | Reasoning, tool use, structured output, function calling |

## Pricing

# Products & Research

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## Models

- **Mercury 2** — general-purpose reasoning
- **Mercury Edit 2** — coding-focused

**Team:** Stanford, UCLA,  
Cornell → Google  
DeepMind, Meta AI,  
OpenAI

## Published Papers

- **Mercury** — arxiv  
2506.17298
- **d1** — diffusion  
reasoning (2504.12216)
- **Block Diffusion** —  
block-level decoding  
(2503.09573)

# Why Text Diffusion Was Hard

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“*Text is discrete — diffusion works naturally on continuous spaces (images).*”

| Images                | Text                         |
|-----------------------|------------------------------|
| Continuous pixels     | Discrete tokens              |
| Natural for denoising | No "partially correct" token |

**Mercury 2's breakthrough:** Making diffusion work on discrete text.

If this scales → **fundamental shift** away from autoregressive dominance.

# QWEN WORLD

## The Simulator

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# What Is Qwen-AgentWorld?

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“ *"A language model trained not to chat or write code — but to **simulate environments**."* ”

Given a **state** + an **action**, it predicts what the environment will return.

**Paper:** arxiv 2606.24597 · **Released:** 2026-06-24 ·  
**License:** Apache 2.0

## Two Model Sizes

• Qwen-AgentWorld-35B-A3B MoE 3B active

# Seven Unified Domains

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A **single model** simulates environments across 7 agent interaction domains:

1.  **MCP** — Model Context Protocol / tool use
2.  **Search** — web/information retrieval
3.  **Terminal** — Linux
5.  **Android** — mobile device interaction
6.  **Web** — browser automation
7.  **OS** — operating system interaction



# Three-Stage Training

| Stage     | What                       | Goal                                                             |
|-----------|----------------------------|------------------------------------------------------------------|
| 1.<br>CPT | Continual Pre-<br>Training | Inject world-modeling capabilities<br>from state-transition data |
| 2.<br>SFT | Supervised Fine-<br>Tuning | Activate next-state-prediction<br>reasoning                      |
| 3. RL     | Reinforcement<br>Learning  | Sharpen fidelity via hybrid rubric +<br>rule rewards             |

Trained on **10M+ real-world interaction trajectories** across all 7 domains.

**Key innovation:** World modeling is the training

# Beats Frontier Models at World Modeling

**AgentWorldBench** — comprehensive benchmark from 5 frontier models on 9 benchmarks:

| Model                     | Score   |
|---------------------------|---------|
| Qwen-AgentWorld-397B-A17B | 58.71 🏆 |
| GPT-5.4                   | 58.25   |
| Claude Opus 4.6           | 57.80   |
| Claude Opus 4.8           | 56.59   |
| Claude Sonnet 4.6         | 56.04   |
| Gemini 2.1 Pro            | 54.57   |

# Two Paradigms for Agent Enhancement

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## 1. Decoupled Environment Simulator

Training in simulated environments  
surpasses training in

## 2. Unified Agent Foundation Model

World-model training

# Why It Matters

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## Today: Reactive Agents

Try action → Observe → Try again

Agents act **blindly** — no ability to think ahead.

## Tomorrow: Planning Agents

Imagine → Evaluate → Choose → Act

With a world model, an agent can **simulate consequences** before execution.

“ *"This is the cognitive leap from reactive to planning*

# SYNTHESIS

**One API · Three Brains**

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# OpenRouter: The Unifying Layer

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|                                                                    |                          |                            |
|--------------------------------------------------------------------|--------------------------|----------------------------|
| OPENROUTER API<br>POST /v1/chat/completions<br>(OpenAI-compatible) |                          |                            |
| Cerebras<br>(fast)                                                 | Mercury 2<br>(diffusion) | Qwen World<br>(simulation) |

**One API key. One endpoint format. Three radically different ways of computing intelligence.**

# The Synthesis Dashboard

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## Cerebras

### Speed Wins

- Time-to-first-token: **sub-200ms**
- Throughput:



## Mercury 2

### Parallel Wins

- Tokens/sec: **1000+**



## Qwen World

### Prediction Wins

- Environment simulation
- Consequence

# The Tradeoffs at a Glance

|                  | Cerebras              | Mercury 2              | Qwen World              |
|------------------|-----------------------|------------------------|-------------------------|
| <b>Core idea</b> | Wafer-scale hardware  | Diffusion for text     | World simulation        |
| <b>Speed</b>     | Sub-second TTFT       | 1000+ tok/s            | N/A (batch)             |
| <b>Paradigm</b>  | Hardware innovation   | Algorithmic innovation | Training innovation     |
| <b>Cost</b>      | Competitive w/ GPU    | \$0.25/\$0.75 per M    | Self-host (open-weight) |
| <b>Best for</b>  | Real-time interaction | High-throughput gen    | Agentic planning        |



# Why Three Architectures, Not One

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“ *"We're entering an era where 'the best model' depends on **what you're building**."* ”

- **Real-time voice agent?** → Cerebras (sub-second latency)
- **Bulk content generation?** → Mercury 2 (1000+ tok/s, cheap)
- **Complex agentic system?** → Qwen World (predict, then act)

The future is **multi-architecture** — and OpenRouter

# The Big Picture

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## Old World

- One architecture (Transformer)
- One training paradigm (autoregressive)
- One deployment pattern (GPU cluster)
- One optimization target

## New World

- **Wafer-scale chips** for instant inference
- **Diffusion models** for parallel generation
- **World models** for consequence prediction
- **Unified APIs** bridging

# Where to Learn More

| Topic             | Links                                                                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cerebras</b>   | <a href="https://cerebras.ai">cerebras.ai</a> · <a href="https://inference.cerebras.ai">inference.cerebras.ai</a>                                                                        |
| <b>Mercury 2</b>  | <a href="https://inceptionlabs.ai">inceptionlabs.ai</a> · <a href="https://platform.inceptionlabs.ai">platform.inceptionlabs.ai</a>                                                      |
| <b>Qwen World</b> | <a href="https://github.com/QwenLM/Qwen-AgentWorld">github.com/QwenLM/Qwen-AgentWorld</a> · <a href="https://qwen.ai/blog">qwen.ai/blog</a>                                              |
| <b>OpenRouter</b> | <a href="https://openrouter.ai">openrouter.ai</a>                                                                                                                                        |
| <b>Papers</b>     | <a href="https://arxiv.org/abs/2506.17298">arxiv: 2506.17298</a> , <a href="https://arxiv.org/abs/2503.09573">2503.09573</a> , <a href="https://arxiv.org/abs/2606.24597">2606.24597</a> |

# Thank You

*Three architectures. One API. The future of AI is multi-paradigm.*